AI Co-Scientist for Target Identification and Macrocyclic Peptide Design

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Design and prototype an AI co-scientist that improves scientific decision-making during an emerging pandemic, specifically for rapid macrocyclic peptide therapeutic discovery under emergency conditions. The system should augment expert decision-making about what to pursue, what to test next, and how to adapt as evidence evolves.

The system takes a poorly characterised pathogen and returns ranked, evidence-backed drug targets; those targets are then designed against using SynX's macrocyclic peptide platform.

Part 1 designs an agentic LLM framework (orchestration layer) to call biodesign tools, and identify and rank targets and epitopes on an unfamiliar pathogen. Part 2 designs non-canonical macrocyclic binders and focused libraries ready for mRNA display selection against the nominated targets, using the SynX platform.

This project does not consider "general" AI scientists (like Google's AI co-scientist, FutureHouse's agents, others) but a narrower setting: emergency peptide discovery, where unknown pathogen biology, timelines of weeks, and limited experimental capacity materially change system design.

Scope

Target identification through to in silico library design. Wet-lab execution, clinical development, manufacturing and regulatory strategy are out of scope for the 10–12 week project.

Core deliverables

1. **System architecture and prototype.** A modular agentic framework covering pathogen intake (genomic sequence), structure prediction, evidence retrieval, target ranking.
2. **Target dossier.** Target identity, structure and confidence, epitope definition, conservation, accessibility, essentiality evidence, target tractability as a reagent, key uncertainties.
3. **Molecule and library design.** Macrocyclic binder designs against a nominated epitope, and associated focused libraries for exploration via the SynX display platform.

4. **Case study**
    a. Apply the system to COVID-19 early-phase data (Jan–Mar 2020): what it would have recommended, when, and how recommendations shift as evidence arrives.
    b. Apply the system against a prototype pathogen.

Stretch goals

- Extension to a second prototype pathogen family as a robustness check
- Extensions and improvements to the SynX macrocyclic peptide design platform
- mRNA display selection round against one designed library (subject to scheduling and time constraints)